

Chapter 2

Background Study and Literature Review

2.1. Introduction

Suicide is a complex and multifaceted public health issue shaped by psychological, social, economic, cultural, and environmental factors. In low- and middle-income countries (LMICs) such as Bangladesh, the challenge is compounded by the near-total absence of a national suicide registry and the pervasive problem of under-reporting, making it extremely difficult to obtain reliable, structured surveillance data. As a result, the most comprehensive data source available to researchers in this context remains newspaper and media reports — a source that is inherently rich in case-level detail yet severely limited by incompleteness and reporting bias [1, 2].

This chapter reviews the existing body of knowledge relevant to this study across four areas. Section 2.2 summarises what is currently known about suicide in Bangladesh based on empirical studies. Section 2.3 analyses the specific challenges associated with newspaper-based suicide datasets, with particular attention to the problem of missing data. Section 2.4 surveys the international literature on missing data imputation methods applied in public health and social research contexts. Section 2.5 reviews related work on unsupervised clustering applied to suicide and mental health data. Section 2.6 synthesises the identified research gaps that motivate the present study, and Section 2.7 provides a brief chapter summary.

2.2. Suicide in Bangladesh: Current Knowledge

Suicide in Bangladesh represents a serious and growing public health burden. Estimates from national health surveys and WHO reports suggest that approximately 10,000 or more people die by suicide annually in the country, though significant under-reporting is widely acknowledged in the literature [1, 2]. A comparative analysis of the Bangladesh Health and Injury Surveys from 2002 and 2015 found that suicide rates increased from 6.2 to 7.7 per 100,000 population over that period, with the highest-risk group shifting from the elderly in 2002 to adolescents by 2015 [16]. These trends underscore the evolving and dynamic nature of the suicide landscape in Bangladesh.

The epidemiology of suicide in Bangladesh exhibits several consistent demographic and methodological patterns across studies. Hanging is by far the most common method,

accounting for approximately 65–82% of completed suicides across different studies [3,17], followed by poisoning, which is especially prevalent in rural areas where pesticides are readily available [2,18]. Females, particularly young married women and female students, are disproportionately represented in suicide statistics — a pattern that reverses the global trend of male predominance [17]. A six-month newspaper content analysis found that 58% of reported cases were female, 24% were students, and 61% were from rural backgrounds, with marital and familial discord being the most frequently cited risk factor [17].

Rural areas consistently show substantially higher suicide rates than urban centres. A population-based study covering approximately 1.16 million people across seven sub-districts found that rural suicide rates were up to 17 times higher than urban rates [2]. Specific districts, including Jhenaidah, Chuadanga, Meherpur, and Narshingdi, have been identified as particularly high-burden areas. A dedicated study of Jhenaidah district recorded a suicide rate of 20.6 per 100,000 — well above both national and global averages — with females exhibiting higher suicide rates than males throughout the study period of 2010–2018 [19].

Among the most vulnerable sub-populations, students have received growing attention. A retrospective analysis of online news portals in Bangladesh found that 984 student suicide cases were identified in the post-pandemic period, with females accounting for 61% of these cases [20]. Contributing factors included academic pressure, board examination timing, family conflict, and failed relationships. COVID-19 further altered suicide patterns, with one study documenting a shift towards higher male incidence during the pandemic period, attributed to economic disruption and job loss among primary breadwinners [21].

Despite this accumulating body of work, the majority of existing Bangladeshi studies share a critical limitation: they rely on manual or semi-automated extraction from newspaper sources and explicitly acknowledge the problem of incomplete records. Variables such as age, exact location, reason for suicide, socioeconomic status, and prior mental health history are frequently absent or inconsistently recorded, severely restricting the depth of analysis that can be conducted [1,3].

2.3. Challenges with Newspaper-Based Suicide Datasets

Newspaper and media-based datasets represent the primary — and often the only — source of case-level suicide data in resource-constrained settings such as Bangladesh. A systematic review of media-based suicide research in LMICs concluded that content analyses of newspaper reports are the dominant research methodology in these settings, driven largely by the absence of alternative data sources and the relatively low cost of media analysis compared to structured epidemiological surveys [22].

However, this reliance comes with well-documented structural limitations. The most critical is the problem of missing data. Reporters covering suicide incidents rarely collect or record the full demographic, socioeconomic, and contextual details that would be required for meaningful quantitative analysis. As a result, critical variables are systematically absent across a large proportion of cases. The following table summarises the key features affected by missingness in newspaper-based datasets, based on consistent patterns reported across the Bangladeshi suicide literature [1–3, 19]:

Feature Category	Typical Missingness Rate	Impact on Analysis
Age / age group	10–25%	Limits age-specific risk profiling
Exact location (lat/long)	40–70%	Prevents geospatial analysis
Reason for suicide	15–40%	Weakens causal inference
Socioeconomic indicators	50–90%	Prevents analysis of poverty and debt links
Weather / environmental data	>60%	Blocks study of seasonal or climatic triggers
Prior mental health history	>90%	Almost impossible to study recurrence

Table 2.1: Typical missingness in newspaper-based suicide datasets in Bangladesh

Beyond missingness, newspaper-based datasets are subject to additional limitations. They are inherently biased towards cases that are deemed newsworthy — typically those involving unusual methods, public locations, or prominent individuals — meaning that common, unremarkable suicides in remote rural areas are systematically under-reported [22]. Reporting inconsistencies in spelling, categorical labels, and temporal formats further complicate data preparation. These challenges force researchers either to perform complete-case analysis — which dramatically reduces sample size and introduces selection bias — or to apply naive imputation strategies such as mean substitution, which introduces distributional bias [5, 6].

2.4. Related Work on Missing Data Imputation in Public Health

The problem of missing data in real-world datasets has generated a substantial methodological literature, particularly within public health and clinical research. Three families of imputation approaches have emerged as dominant in the recent literature: Multiple Imputation by Chained Equations (MICE), K-Nearest Neighbours (KNN) imputation, and ensemble tree-based methods, most notably MissForest.

MICE, originally formalised by Van Buuren and Groothuis-Oudshoorn (2011) [9], operates by iteratively fitting a series of regression models — one per variable — conditioned on all other variables. Its flexibility for mixed-type data (continuous and categorical) and its principled treatment of uncertainty through multiple imputation have made it one of the most widely recommended methods in clinical and epidemiological research. KNN imputation, introduced to the bioinformatics literature by Troyanskaya et al. (2001) [10], fills missing values using the weighted average of the k most similar complete observations. It is computationally efficient and does not impose distributional assumptions, though its performance degrades when datasets are high-dimensional or when local neighbourhood structure is disrupted by high missingness.

MissForest, proposed by Stekhoven and Bühlmann (2012) [11], takes a fundamentally different approach: it trains a separate random forest model for each variable, using all other variables as predictors, and iterates until convergence. This non-parametric, ensemble-based strategy makes it particularly robust to non-linear relationships, interactions, and mixed data types without requiring the specification of explicit regression models. Multiple comparative studies have confirmed MissForest’s superiority over MICE and KNN in medical and biological datasets characterised by moderate to high missingness and complex variable interactions [11, 23, 24].

The following table summarises the most relevant comparative imputation studies, focusing on their domain, methods tested, and key findings:

Study	Domain	Methods Compared	Key Finding
Stekhoven & Bühlmann (2012) [11]	Bioinformatics / mixed-type data	MissForest vs. MICE vs. KNN	MissForest outperforms MICE and KNN on mixed-type data, reducing imputation error by up to 60% in some datasets.
Nijman et al. (2021) [6]	Electronic health records	MICE vs. basic imputation	Multivariate imputation by MICE consistently reduces bias in feature selection compared to simple imputation techniques.
Alkhateeb et al. (2024) [23]	Medical datasets (10 datasets)	MICE, KNN, MissForest, RFE-MF	MissForest consistently outperforms KNN and MICE across diverse medical datasets at missingness rates of 10–50%.
Hamdan et al. (2025) [24]	Healthcare diagnostics	MICE, KNN, MissForest, mean/median	MissForest achieves lowest RMSE and MAE; MICE ranks second; KNN performance is inconsistent across datasets.
Jarrett et al. (2021) [7]	69 real-world tabular datasets	Multiple imputation methods	No single imputation method dominates across all data types and missingness patterns; context-specific evaluation is essential.

Table 2.2: Summary of related work on missing data imputation in public health and related domains

Despite this growing body of comparative work, several important gaps remain. First, the vast majority of imputation benchmarks are conducted on clinical or genomic datasets from high-income settings, and very few have systematically evaluated imputation methods on incomplete social and behavioural datasets from South Asia [7]. Second, most studies evaluate imputation quality in isolation — measuring reconstruction error relative to artificially masked values — rather than assessing the downstream impact of imputation choices on unsupervised learning tasks such as clustering [7, 8]. Third, no prior work has examined the specific problem of predictor selection during imputation — that is, whether using all available columns or only contextually correlated predictors leads to meaningfully different cluster structures — which is one of the novel dimensions explored in the present study.

2.5. Related Work on Clustering in Suicide and Mental Health Research

Unsupervised clustering has been increasingly applied to mental health and suicide data as a means of identifying latent subgroups within heterogeneous populations. Unlike supervised approaches that require labelled outcomes, clustering allows the data itself to reveal structure — a particularly appropriate strategy when, as in the present study, no ground-truth class labels exist.

At the broadest level, Orchard et al. (2024) [25] applied K-Means and hierarchical clustering to a large Canadian community health survey, identifying four distinct mental health profiles with meaningfully different healthcare service-use patterns. Their work demonstrated that unsupervised methods can reveal clinically and policy-relevant subgroups in population-level mental health data that would not be apparent through traditional regression approaches.

More directly relevant to the suicide context, Xiao et al. (2025) [26] published what is described as the first study to apply unsupervised machine learning to a comprehensive set of social determinants of health specifically for suicide risk across U.S. communities. Using K-Means clustering on large population-level data, they identified geographically coherent community-level risk clusters — including one characterised by social isolation and poor healthcare access, and another by economic distress and family instability — each requiring distinct prevention strategies. This work closely parallels the motivations of the present study, though the U.S. context, the availability of structured administrative data, and the focus on community-level rather than individual-level variables make the methodological challenges substantially different from those encountered with incomplete Bangladeshi newspaper data.

Kim et al. (2025) [27] applied agglomerative nesting followed by K-Means to psychological autopsy data from 391 suicide deaths in Spain, identifying three impulsivity-based subprofiles. Their study highlights the significant heterogeneity among individuals who die by suicide — a finding that directly motivates the use of clustering over aggregate-level analyses. Ong et al. (2022) [28] used cluster analysis on stressor and risk factor data from student suicides in Hong Kong, identifying four meaningful profiles (school distress, hidden, family and relationship, and numerous issues), each with distinct prevention implications.

In the Bangladeshi context specifically, prior studies have not applied any form of algorithmic clustering to suicide data. Arafat et al. (2018) [3], one of the most cited newspaper-based analyses of Bangladeshi suicides, explicitly acknowledged that high levels of missingness in the dataset precluded any systematic subgroup analysis beyond basic cross-tabulation. This observation directly identifies the methodological gap that the present study addresses.

The following table summarises the most relevant clustering studies in the suicide and mental health domain:

Study	Country / Data	Clustering Approach	Main Finding
Orchard et al. (2024) [25]	Canada (community health survey)	K-Means, hierarchical clustering	Four distinct mental health profiles identified, each associated with different healthcare service-use patterns.
Xiao et al. (2025) [26]	USA (social determinants data)	Unsupervised machine learning (K-Means)	First study to use unsupervised ML on social determinants of health for suicide; community-level risk clusters identified.
Kim et al. (2025) [27]	Spain (psychological autopsy, 391 cases)	Agglomerative nesting + K-Means	Three impulsivity-based suicide profiles identified; highlights significant heterogeneity among suicide decedents.
Ong et al. (2022) [28]	Hong Kong (student suicides)	Cluster analysis on stressors and risk factors	Four student suicide profiles found (school distress, hidden, family/relationship, numerous issues); supports differentiated interventions.
Arafat et al. (2018) [3]	Bangladesh (newspaper data)	Manual grouping only	No algorithmic clustering applied; missingness in newspaper data cited as primary barrier to advanced analysis.

Table 2.3: Summary of related work on clustering in suicide and mental health research

A critical observation across all reviewed clustering studies is that none of them systematically varied both the imputation strategy and the clustering algorithm to understand how imputation choices affect downstream cluster quality. Furthermore, existing studies have almost universally relied on standard internal validity indices — Silhouette score, Calinski–Harabasz, and Davies–Bouldin — without accounting for the practical problem of cluster size imbalance [29,30]. When one cluster contains the vast majority of observations and another contains only a handful of outliers, standard validity indices can still report high scores, making the solution appear better than it is in practice. This limitation is particularly acute in real-world social datasets where natural group imbalance is the norm rather than the exception.

2.6. Problem Analysis and Research Gap

From the foregoing review, the following critical gaps emerge that directly motivate the design of the present study:

Gap 1 — No large-scale, enriched suicide dataset for Bangladesh: The largest existing newspaper-based Bangladeshi suicide datasets contain at most a few hundred records with limited attributes and cover single years. No publicly available dataset systematically covers multiple years (2017–2025) or includes socioeconomic, clinical, and environmental attributes alongside the basic demographic information.

Gap 2 — No systematic imputation benchmark for this data type: While the international imputation literature is extensive, no study has systematically benchmarked state-of-the-art imputation methods (MICE, KNN, MissForest) on a large, real-world, newspaper-derived social dataset from South Asia with the extreme missingness levels characteristic of this domain (50–90% for several key variables).

Gap 3 — No multi-method clustering study on Bangladeshi suicide data: No published work has applied multiple clustering algorithms to Bangladeshi suicide records, let alone examined how the choice of imputation method influences the resulting cluster structure. The only existing clustering applications in this domain have been conducted on Western datasets with structured, near-complete data.

Gap 4 — No evaluation framework accounting for cluster imbalance: Standard internal clustering validity indices do not penalise solutions where a single cluster dominates the data or where density-based algorithms classify the majority of points as noise. For naturally imbalanced social datasets, a fairness-aware evaluation component is needed to distinguish practically useful clusterings from statistically artefactual ones.

Gap 5 — No consideration of predictor selection effects during imputation: Existing comparative imputation studies rarely examine how the choice of predictor set — using all available columns versus only contextually correlated ones — affects both imputation quality and downstream clustering outcomes. This dimension is entirely unexplored in the Bangladeshi public health context.

The present study directly addresses all five gaps by: (1) expanding the publicly available dataset from 760 to 1,617 cases with 9 new attributes covering 2017–2025; (2) implementing an exhaustive 32-pipeline comparison of three imputation methods under two predictor strategies and four clustering algorithms; (3) introducing a composite scoring framework that incorporates a novel balance metric alongside five standard validity indices; and (4) providing fully interpretable, actionable cluster profiles grounded in the Bangladeshi social context.

Dimension	Arafat et al. 2018 [3]	Xiao et al. 2025 [26]	Kim et al. 2025 [27]	Jarrett et al. 2021 [7]	This Study
Country / Context	Bangladesh	USA	Spain	69 datasets	Bangladesh
Data source	Newspaper	Administrative	Clinical autopsy	Various	Newspaper
Dataset size	~360 cases	Large population	391 cases	Varies	1,617 cases
Missing data handling	None / complete case	Not applicable	Not applicable	Imputation benchmarks	MICE + KNN + MissForest $\times$ 2 strategies
Clustering applied	None	K-Means only	Agglomerative + K-Means	None	4 algorithms $\times$ 8 datasets = 32 pipelines
Evaluation metric	Manual grouping	Standard metrics	Standard metrics	RMSE / reconstruction	5 standard + novel balance metric
Multi-year coverage	2009–2018	Multiple years	Single study	N/A	2017–2025
Reproducible pipeline	No	Partial	No	Yes	Yes (12 notebooks, open source)
Novel metric proposed	No	No	No	No	Yes (balance metric)

2.7. Summary

This chapter has established that suicide in Bangladesh is a serious and growing public health concern, with consistent demographic patterns emerging across two decades of newspaper-based research — including the predominance of hanging as method, over-representation of young females and students, sharp rural–urban disparities, and high-burden districts such as Jhenaidah and Chuadanga. It has also demonstrated that newspaper data remains the only comprehensive source of case-level suicide information in Bangladesh, but that this source is characterised by extreme and non-random missingness that fundamentally limits the analytical methods that can be applied.

The international literature on missing data imputation provides a strong methodological foundation: MissForest consistently outperforms MICE and KNN on mixed-type data with moderate to high missingness, though no benchmark has been conducted in the specific context of newspaper-based social data from South Asia. The clustering literature confirms that unsupervised methods can identify meaningful, intervention-relevant subgroups from mental health data, but no such analysis has been attempted on Bangladeshi suicide records, and no existing study has jointly optimised imputation and clustering or evaluated solutions with a fairness-aware metric.

The identified research gaps collectively justify the methodological focus of the present study. Chapter 3 describes the comprehensive research framework designed to address these gaps.